INTRODUCTION

Galaxy rotation curves remain a central test for theories of gravity and dark matter. Two paradigms dominate:

- **ΛCDM with NFW halos**: excellent per-galaxy fits with 4 free parameters ($\Upsilon_{\rm disk}$, $\Upsilon_{\rm bulge}$, M_{200}, c), but halo parameters vary widely across galaxies [7].
- **MOND**: a universal acceleration threshold $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2} with only 2 per-galaxy parameters ($\Upsilon_{\rm disk}$, $\Upsilon_{\rm bulge}$), but systematically poorer fits for some galaxy types [1, 6].

The Radial Acceleration Relation [RAR; 3, 4] demonstrates a tight empirical correlation between baryonic and observed accelerations, suggesting a universal transition between regimes. However, standard frameworks cannot simultaneously optimize for fit quality and parameter universality.

We ask: *does the hypothesis space contain families that occupy the trade-off region between NFW's fit quality and MOND's universality?*

DATA

We use the SPARC dataset [2]: 175 late-type galaxies with Spitzer 3.6 μm photometry and high-quality rotation curves. After filtering for quality flags $Q \leq 2$ and minimum 5 data points: 159 galaxies.

SPARC provides pre-computed velocity contributions ($V_{\rm disk}$, $V_{\rm bulge}$ at $M/L = 1$, $V_{\rm gas}$) at each radius, enabling mass model construction without analytic profile assumptions. The split is deterministic: 99 train, 30 validation, 30 test, stratified by luminosity.

For modified gravity laws, enclosed baryonic mass is derived as:

$$M_{\rm bary}(r) = \frac{r}{G}\left(V_{\rm gas}^2 + \Upsilon_d V_{\rm disk}^2 + \Upsilon_b V_{\rm bul}^2\right) \qquad (1)$$

where $\Upsilon \in [0.1, 1.5]$ $M_\odot/L_\odot$ at 3.6 μm.

METHOD

Law Families

Each family defines an acceleration function $a(r, M_{\rm enc}, \rho)$. The three baselines are Newton baryons-only ($a = GM/r^2$, 2 params), Newton+NFW (standard halo, 4 params), and simple MOND ($a = a_N \nu(a_N/a_0)$, 2 params with a_0 fixed).

The generated family is PiecewiseRegime:

$$a(r) = a_0\left[\sigma(x)\left(\frac{a_N}{a_0}\right)^{\alpha_{\rm in}} + (1-\sigma(x))\left(\frac{a_N}{a_0}\right)^{\alpha_{\rm out}}\right] \qquad (2)$$

where $a_N = GM_{\rm bary}/r^2$, $\sigma(x) = (1 + e^{-x})^{-1}$, $x = \log_{10}(a_N/a_0)/w$. All terms inside brackets are dimensionless; a_0 provides acceleration units. This family nests both Newton ($\alpha_{\rm in} = \alpha_{\rm out} = 1$) and deep-MOND ($\alpha_{\rm in} = 1$, $\alpha_{\rm out} = 0.5$) as special cases.

Four-Dimensional Fitness

Each family is scored on four independent axes (higher = better):

F1 (fit quality): $= 1/(1 + \text{median}\ \chi^2_{\rm red})$.

F2 (universality): $= 1/(1 + \overline{\rm CV}_j)$, where $\text{CV}_j = \text{IQR}(\theta_j)/|\text{median}(\theta_j)|$ for each law parameter θ_j. Families with no law parameters (Newton baryons, MOND) get $F2 = 1$.

F3 (simplicity): $= 1/(k_{\rm law} + k_{\rm galaxy} + S)$, where S counts mathematical operations.

F4 (RAR transfer): $= 1/(1 + \text{RMS}\ \log_{10}(g_{\rm pred}/g_{\rm obs}))$.

No weighted sum. **Pareto dominance** determines optimal families: A dominates B iff $A \geq B$ on all axes and $A > B$ on at least one.

Mutation Operators

M4 (Freezing): Fix one law parameter at its cross-galaxy median.

M2 (Correction): Add multiplicative term $(1 + c_1 R^{p_1})$ with dimensionless ratio R.

M3 (Composition): Blend two families via sigmoid transition.

All mutations preserve dimensional consistency. Population capped at 30 families; structural complexity $S \leq 15$.

RESULTS

Baseline Comparison

Under BIC, Newton+NFW wins (median BIC = 26.2, 43/99 wins). PiecewiseRegime's best-fit law parameters converge to $\alpha_{\rm in} = 1.05$, $\alpha_{\rm out} = 0.52$, recovering MOND-like exponents without prior knowledge.

Pareto Front and the Gap

On the 4D fitness, all four baselines are Pareto-optimal but no family achieves $F1 \geq 0.35$ and $F2 \geq 0.70$ simultaneously—a gap between NFW's fit quality and MOND's universality.

After two generations of evolution, **PR-only-a_0-free**—PiecewiseRegime with $\alpha_{\rm in} = 1.05$, $\alpha_{\rm out} = 0.52$, $w = 0.615$ frozen, only a_0 varying per galaxy—fills this gap:

TABLE I. Test set 4D fitness (30 galaxies)

Family	F1	F2	F3	F4
Newton baryons	0.023	1.000	0.500	0.738
Newton+NFW	0.520	0.479	0.200	0.907
MOND-simple	0.261	1.000	0.250	0.886
PiecewiseRegime	0.424	0.575	0.077	0.908
PR-only-a_0	**0.389**	**0.915**	**0.100**	**0.893**

PR-only-a_0-free achieves $\sim$85% of NFW's fit quality with nearly double the universality ($F2 = 0.92$ vs. 0.48) and one fewer per-galaxy parameter (3 vs. 4).

What Determines a_0?

Correlation of per-galaxy $\log_{10} a_0$ against six galaxy properties (159 galaxies): the strongest correlator is $V_{\rm max}$ ($r = 0.36$, $p = 2.4 \times 10^{-6}$), explaining only 13% of variance. No property exceeds $r = 0.4$. The a_0 scatter is 0.33 dex, largely independent of observed properties.

DISCUSSION

The universal transition shape ($\alpha_{\rm in} \approx 1.05$, $\alpha_{\rm out} \approx 0.52$, $w \approx 0.62$) is consistent with known RAR phenomenology [3, 4]. The variable-a_0 question connects to the debate between **(author?)** [8] and **(author?)** [5]. Our contribution is not to this debate but to the methodology.

Limitations. (1) Single dataset (SPARC); no lensing or cluster constraints. (2) Mass-to-light degeneracy, though equal for all families. (3) PR-only-a_0-free is phenomenological, not derived from a field theory. (4) The a_0 variation may reflect observational systematics.

What we cannot claim: discovery of new physics, evidence for or against dark matter, or that a_0 is a varying fundamental constant.

CONCLUSION

A multi-objective evolution engine identifies a Pareto-optimal law family combining NFW-level fit quality with MOND-level universality on 159 SPARC galaxies. The transition shape is universal; only the scale varies. The contribution is the method—multi-objective Pareto comparison of hypothesis families—not the astrophysical phenomenology, which is consistent with existing literature.

Code and data: `https://github.com/SaulVanCode/galaxy-rotation-worlds`

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